

Final Exam – Medicinal Plants (Very Hard – Version B)

Essay-Based Examination

Syllabus: Lectures 5–12

Time: 3 Hours

Section A – Very Hard Essay Questions (Version B)

1. A patient with chronic constipation develops dependency on herbal laxatives. Discuss the pathophysiology of laxative abuse using Senna as an example, and outline possible complications.
2. Explain the biochemical basis for the antibacterial activity of Uva Ursi and discuss why urinary pH is critical for its effectiveness.
3. Compare Boldo and Artichoke in terms of therapeutic uses, active constituents, and toxicological concerns.
4. Discuss the role of bile secretion in digestion and lipid metabolism, illustrating your answer with suitable medicinal plants.
5. Evaluate the clinical use of Guava leaves in gastrointestinal disorders, explaining the mechanism of action.
6. Analyze the pharmacological basis of Ivy leaves as expectorants and explain why they are not recommended in young children.
7. Describe the mechanism of action of Eucalyptus oil in respiratory tract disorders and discuss the risks of systemic absorption.
8. Discuss the stimulant effects of Tea leaves, including benefits and adverse effects associated with excessive consumption.
9. Describe the mechanism of anticholinergic toxicity produced by Hyoscyamus and explain its clinical manifestations.
10. Discuss the therapeutic applications of Jaborandi, emphasizing the pharmacology of pilocarpine.
11. Explain the pharmacological actions of Coca leaves and discuss ethical and clinical concerns related to their misuse.
12. Analyze Digitalis therapy in heart failure, explaining both therapeutic effects and factors leading to toxicity.

13. Discuss electrolyte imbalance as a major determinant of *Digitalis* toxicity.
14. Explain the role of Ginkgo in neurovascular disorders and discuss potential drug interactions.
15. Discuss the importance of taxonomy in differentiating medicinal plants from toxic adulterants.
16. Compare monocotyledonous and dicotyledonous plants with respect to diagnostic botanical characters.
17. Explain how improper plant identification can lead to serious clinical consequences.
18. Discuss the pharmacological significance of saponins in medicinal plants, with examples.
19. Evaluate the risks associated with long-term use of stimulant laxatives derived from plants.
20. Differentiate between hepatoprotective, choloretic, and cholagogue actions, giving suitable examples.

Section B – Model Answers

1. Chronic use of Senna causes colonic nerve damage, dependency, and electrolyte imbalance, especially hypokalemia.
2. Arbutin is hydrolyzed in alkaline urine to hydroquinone which exerts antibacterial action.
3. Boldo contains ascaridole causing toxicity, while Artichoke contains cynarin with choleretic effect.
4. Bile emulsifies fats and aids cholesterol excretion; Artichoke enhances bile secretion.
5. Guava leaves contain tannins that exert astringent effect reducing intestinal secretions.
6. Ivy contains saponins that liquefy mucus but may cause GI irritation in children.
7. Eucalyptus oil acts as decongestant but systemic absorption may cause CNS depression.
8. Tea contains caffeine causing stimulation but excessive intake leads to insomnia and palpitations.
9. Hyoscyamus blocks muscarinic receptors causing anticholinergic syndrome.
10. Jaborandi contains pilocarpine, a muscarinic agonist used in glaucoma.
11. Coca leaves stimulate CNS but misuse leads to addiction and cardiovascular toxicity.
12. Digitalis increases myocardial contractility but has narrow therapeutic index.
13. Hypokalemia enhances Digitalis binding to Na^+/K^+ -ATPase increasing toxicity.
14. Ginkgo improves cerebral blood flow but increases bleeding risk.
15. Taxonomy ensures accurate identification preventing toxic substitution.
16. Monocots have one cotyledon; dicots have two with reticulate venation.
17. Misidentification may result in toxicity or therapeutic failure.
18. Saponins reduce surface tension of mucus aiding expectoration.
19. Long-term stimulant laxatives cause dependency and electrolyte imbalance.
20. Hepatoprotective protects liver cells; choleretic increases bile secretion; cholagogue promotes bile flow.